HW 2.1: Having studied the ground state energy of the Hydrogen atom with the variational method, we extend it to some more interesting systems. We will estimate the ground state energy of atoms with more than one electron using the variational method. Consider an atom with nuclear charge Z and two electrons. The Hamiltonian of this system is given by

$$H(x) = -\frac{\hbar^2}{2m}\left(\nabla_1^2 + \nabla_2^2\right) - \frac{Ze^2}{4\pi\epsilon_0}\left(\frac{1}{r_1} + \frac{1}{r_2}\right) + \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|}, \tag{1}$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the position vectors of the two electrons, and $r_1 = |\mathbf{r}_1|$, $r_2 = |\mathbf{r}_2|$. The first term represents the kinetic energy of the electrons, the second term represents the attraction between the electrons and the nucleus, and the last term represents the repulsion between the two electrons. Using the variational method, we can estimate the ground state energy of this system by choosing a trial wavefunction of the form

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{Z_{\text{eff}}^3}{\pi a_0^3} e^{-Z_{\text{eff}}(r_1 + r_2)/a_0}, \tag{2}$$

where Z_{eff} is an effective nuclear charge that we will vary to minimize the energy expectation value, and a_0 is the Bohr radius ($a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2}$).

a) Calculate the expectation value of the Hamiltonian $\langle \hat{H} \rangle$ using the trial wavefunction $\psi(\mathbf{r}_1, \mathbf{r}_2)$, and find the value of Z_{eff} that minimizes this expectation value.

b) Use this value of Z_{eff} to estimate the ground state energy of the atom for $Z = 1$ (H^-), $Z = 2$ (He), and $Z = 3$ (Li^+) in eV.

c) Is an anionic hydrogen atom (H^-) possible, i.e., is it a bound state?

d) What about for a muonic atom ($m \simeq m_\mu \simeq 200 \times m_e$)? Can a muonic hydrogen anion ($2\mu^- p$) exist as a bound state?

You may find the following integrals useful:

$$\begin{aligned}
\int_0^\infty dr\, r\, e^{-2Z_{\text{eff}} r/a_0} &= \frac{a_0^2}{4Z_{\text{eff}}^2}, \\
\int_0^\infty dr\, r^2\, e^{-2Z_{\text{eff}} r/a_0} &= \frac{a_0^3}{4Z_{\text{eff}}^3}. \\
\int d^3 r_1 \int d^3 r_2\, \frac{e^{-2Z_{\text{eff}}(r_1 + r_2)/a_0}}{|\mathbf{r}_1 - \mathbf{r}_2|} &= \frac{5\pi^2 a_0^5}{8Z_{\text{eff}}^5}.
\end{aligned} \tag{3}$$

HW 2.2: Variational Method for Excited States

a) Show that the variational method can be used to estimate the energy of the first excited state of a quantum system, namely, prove the following upper bound to the energy of the first excited state

$$E_1 \leq \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle}, \tag{4}$$

where ψ_0 is the ground state wavefunction and ψ is a trial wavefunction that respects $\langle \psi_0 | \psi \rangle = 0$ (you can assume that we know the ground state $|\psi_0\rangle$ and ground energy E_0 already and that the energy eigenvalues are non-degenerate).

b) **(Bonus)** Apply this generalized variational method to the quantum harmonic oscillator to estimate the energy of the first excited state. Use a trial wavefunction of the form

$$\psi(x) = x e^{-\alpha x^2}, \tag{5}$$

where α is a variational parameter. Calculate the expectation value of the Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2} m\omega^2 x^2, \tag{6}$$

and find the value of α that minimizes this expectation value. Compare your result with the exact energy of the first excited state, $E_1 = \frac{3}{2}\hbar\omega$.